R-Select: A Robust Multi-Metric Data Selection Approach for Fine-Tuning Large Language Models

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Abstract

The transition from architecture-centric scaling to data-centric refinement has established high-quality data as a critical determinant of Large Language Model performance, particularly for complex reasoning and instruction following. However, effective data selection remains a persistent bottleneck: simple heuristic filters often fail to capture multifaceted data features (e.g., reasoning depth and diversity), while advanced model-based scoring methods typically prioritize isolated quality dimensions, failing to provide a holistic assessment. To address these challenges, we propose R-Select, a robust and scalable framework that optimizes data selection with 30 distinct quality metrics. Recognizing that optimizing such a high-dimensional feature space is non-trivial, R-Select introduces a novel hierarchical optimization strategy. This approach structurally decomposes the search problem by first clustering correlated metrics into functional groups based on statistical dependencies. It then executes a two-stage optimization process: performing *intra-group refinement* to maximize local representational power, followed by *inter-group integration* to balance global quality domains. Crucially, to ensure computational efficiency, R-Select employs a low-resource proxy strategy, utilizing a lightweight model on a small data subset to learn an optimal selection policy that is transferable to the target model. Extensive experiments demonstrate that R-Select consistently outperforms both heuristic baselines and model-based methods, offering a robust solution for high-quality data curation. Our code is available at https://github.com/OpenDataArena/R-Select.

CCS Concepts

• **Information systems → Data mining**; • **Computing methodologies → Natural language processing**; **Supervised learning**.

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Supervised fine-tuning, Data selection, Large language models

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1 Introduction

The evolution of Large Language Models (LLMs) has marked a paradigmatic shift from architecture-centric scaling [34] to data-centric refinement [69]. It is now widely acknowledged that the capabilities of LLMs, particularly in reasoning and instruction following, are fundamentally shaped by the quality and composition of data during the post-training stage [58, 93]. Therefore, how to select high-quality data has emerged as a critical research topic for guaranteeing model performance.

However, defining and quantifying *data quality* remains a persistent challenge. Existing approaches largely fall into two extremes: heuristic filtering and model-based scoring. Heuristic methods typically rely on isolated, single-dimensional indicators [59, 61] (e.g., entropy or token length), which effectively filter basic noise but fail to capture the multifaceted nature of high-quality data. Conversely, recent model-based approaches (e.g., Instruction-following difficulty [40] or QuRating [76]) offer sophisticated assessments but often prioritize specific quality facets in isolation. Consequently, they struggle to capture the full spectrum of data value, failing to address the multifaceted requirements essential for comprehensive model development. There is a distinct absence of a framework that is both comprehensive in covering diverse quality dimensions and systematic in optimizing their integration.

Constructing such a framework presents a significant optimization bottleneck. To holistically evaluate data, one might introduce multiple indicators capturing distinct dimensions like diversity and difficulty [23, 86]. However, integrating these heterogeneous

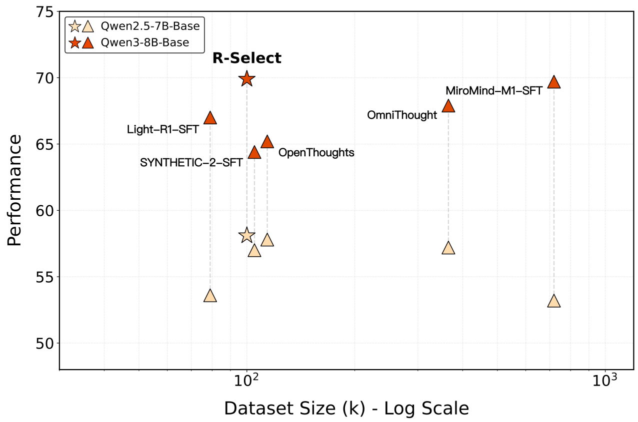

Figure 1: Performance vs. dataset size. R-Select shows state-of-art result on model SFT performance with high efficiency.

metrics into a unified selection policy creates a high-dimensional combinatorial search space. In this regime, performing a direct search over the full hyperparameter space is computationally prohibitive, as standard optimization algorithms struggle to converge efficiently due to the curse of dimensionality [4]. Moreover, manual tuning based on empirical intuition is no longer feasible given the complex trade-offs between conflicting quality objectives.

To address these challenges, we propose R-Select, a robust and automated data selection framework. As illustrated in Figure 1, R-Select achieves competitive performance and data efficiency compared to existing baselines. R-Select formulates data selection as a structured weight-optimization problem. To capture the granular nuances of high-quality data, we construct a comprehensive feature space comprising dozens of distinct quality indicators—spanning semantic diversity, reasoning complexity, and general linguistic features—aggregated into a unified linear scoring function. This design choice prioritizes coverage and interpretability, allowing us to explicitly quantify the contribution of each dimension to the final model performance.

The core innovation of R-Select lies in its hierarchical optimization strategy synergized with a low-resource proxy mechanism. To circumvent the prohibitive computational costs of iterative validation, we employ a lightweight model and reduced data subsets as cost-effective surrogates, enabling rapid feedback loops for weight optimization. Structurally, to overcome the efficiency bottlenecks, we decompose the problem based on metric semantics with (1) *Intra-group refinement*: we first cluster indicators into functional groups based on statistical correlations. Within each group, we optimize for a local weight configuration that maximizes the representational power of that specific quality facet; (2) *Inter-group integration*: we then freeze the intra-group structures and optimize the global weights across these functional domains.

This divide-and-conquer approach transforms an intractable high-dimensional problem into a series of stable, low-dimensional sub-problems, ensuring both convergence efficiency and optimization robustness. Our contributions are summarized as follows:

- We introduce a multi-dimensional evaluation system that integrates 30 indicators, moving beyond heuristics to capture a comprehensive view of data quality.
- We propose a hierarchical optimization framework that automates the configuration of quality weights. By decoupling intra-group refinement from inter-group integration, we achieve stable optimization in high-dimensional spaces.
- Extensive experiments demonstrate that R-Select outperforms both heuristic baselines and complex model-based selectors. Furthermore, we analyze the optimized weights to provide interpretable insights into what constitutes *high-quality* data for different downstream tasks.

2 Related Work

2.1 From Scale to Quality in SFT Datasets

Early supervised fine-tuning (SFT) efforts were primarily driven by scale. Datasets such as Alpaca [68], ShareGPT [8], and early versions of OpenAssistant [36] were largely constructed by aggregating massive amounts of user interactions or model-generated responses, with limited control over quality and consistency. While these corpora significantly advanced instruction-following capabilities, their high heterogeneity also introduced substantial noise, redundancy, and stylistic biases. A second wave of work subsequently shifted toward more curated and structured SFT data construction. Representative datasets including FLAN [46], Orca[54], and Tulu [37] place greater emphasis on instruction diversity, explicit reasoning traces, and broad domain coverage rather than sheer data volume. More recent resources such as WizardLM [79]and UltraInteract [85], and Math-Shepherd [72] further prioritize multi-step reasoning, verifiable solution structures, and fine-grained difficulty control, reflecting a growing consensus that data composition is as critical as data scale. However, even within these carefully curated large-scale mixtures, a considerable proportion of low-quality or misaligned samples remains. Prior studies including LIMA [93], LIMO [82], ODA-Mixture[24] consistently show that carefully filtered small subsets can outperform much larger but less refined corpora. These findings suggest that simply "scaling up" data is neither sufficient nor always beneficial, and can even be detrimental without explicit quality control. Taken together, the SFT data paradigm is gradually shifting from "more data" to "better data," with increasing emphasis on systematic data evaluation, filtering, and optimization to improve sample efficiency and model generalization.

2.2 Data Selection for SFT

Existing approaches for SFT data selection can be broadly categorized into three paradigms: model-based evaluation, heuristic-based evaluation, and LLM-as-Judge evaluation. Model-based methods estimate sample value using learned models and signals derived from both training and inference. In addition to widely used reward models (e.g., Skywork-Reward-V2 [42], INF-ORM [53]), recent work has developed more targeted scorers. For example, DEITA [45] leverages LLMs to rewrite and generate data with systematically varied quality in order to train a dedicated complexity scorer. Complementarily, IFD [40] quantifies how hard an instruction is to follow, while UPD [87] measures uncertainty-based difficulty using Shannon entropy and loss statistics. Along a different dimension, another line of work [14, 35, 38] employs gradient-based metrics to estimate the importance of individual training samples. Finally, Instag [47] trains a tagging model to assess SFT data quality and diversity via interpretable tags. Heuristic methods rely on computationally cheap, rule-based features. Beyond basic indicators such as token length, entropy, and unique-gram ratio, lexical diversity metrics including MTLD, HHD, and Vocd [52] are commonly used to characterize linguistic richness. In practice, researchers [95] also design regular-expression rules and format checks to ensure basic structural correctness. At the dataset level, deduplication via hashing and MinHash remains a standard preprocessing step. LLM-as-Judge methods prompt strong models to assess semantic quality. Early work such as AlpaGasus [9] pioneered direct LLM scoring of data accuracy, while ASK-LLM [64] derived continuous scores from conditional probabilities. More recently, GRA [23] proposed a review–adjudication framework that aggregates judgments from multiple LLMs to improve evaluation reliability. SelectIT [88] introduces uncertainty-aware scoring by varying prompt phrasing, analyzing token distributions, and aggregating across models

Overall, while these approaches provide valuable signals, most prior work relies on single-metric ranking or manually designed aggregation rules. In contrast, R-Select formulates data selection as a continuous multi-metric optimization problem that learns to combine heterogeneous signals automatically within a unified framework, rather than treating them in isolation.

3 Preliminaries

We first establish the mathematical notations and formally define the data selection task within a bi-level optimization framework.

3.1 Notations and Definitions

Data and Features. Let $\mathcal{D} = \{d_j\}_{j=1}^{M}$ be the candidate pool of M samples. For each sample $d \in \mathcal{D}$, we assume a feature extraction process that characterizes its quality across F distinct dimensions. Specifically, we denote $f_i(d)$ as the scalar value capturing the i-th specific quality aspect (e.g., semantic complexity, logical correctness, or lexical diversity). These feature values are concatenated into a raw quality vector $\mathbf{f}(d) = [f_1(d), \dots, f_F(d)]^\top \in \mathbb{R}^F$. To mitigate scale variance across heterogeneous metrics, we define $\bar{\mathbf{f}}(d) = \phi(\mathbf{f}(d)) \in [0,1]^F$ as the normalized feature representation, where $\phi(\cdot)$ is a pre-defined normalization operator.

Selection Mechanism. We introduce a learnable weight vector $\mathbf{w} \in [0,1]^F$, where each component w_i captures the relative importance of the i-th quality dimension. The composite quality score for a sample d is computed via the linear projection $S(d, \mathbf{w}) = \mathbf{w}^\top \bar{\mathbf{f}}(d)$. Given a selection budget $N \ll M$, we define a selection indicator $\mathcal{I}(d, \mathbf{w}) \in \{0, 1\}$, such that $\mathcal{I}(d, \mathbf{w}) = 1$ if $\mathrm{rank}(S(d, \mathbf{w})) \le N$, and 0 otherwise, where $\mathrm{rank}(\cdot)$ denotes descending rank among all samples. The resulting parameterized subset is denoted as $\mathcal{D}_{\mathrm{sel}}(\mathbf{w}) = \{d \in \mathcal{D} \mid \mathcal{I}(d, \mathbf{w}) = 1\}$.

3.2 Problem Formulation

Our objective is to identify an optimal weight vector $\mathbf{w}^*$ that maximizes the generalization capability of the model by curating the most effective training subset. We formulate this task as a bi-level optimization problem:

$$\min_{\mathbf{w}} \quad \mathcal{J}(\mathbf{w}) = \mathcal{L}_{\mathrm{val}}(\theta^*(\mathbf{w}); \mathcal{D}_{\mathrm{val}}) \tag{1}$$

$$\text{s.t.} \quad \theta^*(\mathbf{w}) = \arg\min_{\theta} \mathcal{L}_{\mathrm{train}}(\theta; \mathcal{D}_{\mathrm{sel}}(\mathbf{w})) \tag{2}$$

where θ represents the parameters of LLM, $\mathcal{D}_{\mathrm{val}}$ denotes a held-out gold-standard validation set, and $\mathcal{L}_{\mathrm{train}}$, $\mathcal{L}_{\mathrm{val}}$ are the standard cross-entropy losses for supervised fine-tuning.

In this framework, the inner optimization (Eq. 2) corresponds to the model training phase on the selected subset, while the outer optimization (Eq. 1) seeks to refine the data-curation policy $\mathbf{w}$ based on performance feedback. Notably, the discrete nature of the selection indicator $\mathcal{I}$ makes the outer objective $\mathcal{J}(\mathbf{w})$ non-differentiable with respect to $\mathbf{w}$, thereby necessitating the use of derivative-free optimization methods, which we introduce in Sec. 4.

4 Hierarchical Bayesian Data Selection

This section presents R-Select, a hierarchical Bayesian optimization framework that synergizes heterogeneous quality metrics for automated SFT data selection. By structuring the search space through metric clustering, R-Select transforms a high-dimensional, noisy selection problem into a tractable coarse-to-fine optimization process. The overall architecture is illustrated in Fig. 2.

4.1 Search Space Structuring

R-Select operates on a comprehensive suite of quality metrics spanning three complementary categories: (i) model-based metrics (e.g., IFD [40] and UPD [86]), (ii) heuristic metrics (e.g., Length [78] and MTLD [52]), and (iii) LLM-as-Judge metrics [6]. A complete list of all metrics and their formal definitions is provided in Appendix C. To transform these raw signals into a representation that facilitates efficient optimization, we first implement a standardization pipeline: (i) **Outlier Mitigation:** anomalous samples with extreme metric values are filtered to prevent skewing the weight distribution; (ii) **Range Normalization:** Min-Max scaling is applied to map all metrics to $[0, 1]$; and (iii) **Semantic Alignment:** for metrics where lower values denote superior quality (e.g., perplexity), we apply score inversion $f'(d) = 1 - f(d)$, ensuring a consistent "higher-is-better" semantics across the feature vector $\bar{\mathbf{f}}(d)$.

Directly optimizing the weight vector $\mathbf{w}$ in the raw F-dimensional space often encounters the *curse of dimensionality* and a highly non-convex landscape, leading to slow convergence. To mitigate this, we introduce a hierarchical search space based on Ward's Hierarchical Clustering [55]. Specifically, we define the distance between any two metrics i and j as $\mathrm{dist}_{ij} = 1 - |\rho_{ij}|$, where ρ_{ij} denotes the pairwise Pearson correlation coefficient. We then recursively group metrics into K coherent clusters $\{C_k\}_{k=1}^{K}$ by minimizing intra-cluster variance.

This hierarchical design serves a dual purpose. First, it enables **dimensionality reduction** by allowing the optimizer to prioritize cluster-level importance before refining individual metric weights. Second, it facilitates **variance reduction**; since many metrics capture redundant quality signals, aggregating correlated features acts as a form of structural regularization, yielding more robust and interpretable signals than raw, noisy individual metrics. This transformation effectively converts a prohibitively irregular search space into a structured domain for efficient Bayesian exploration.

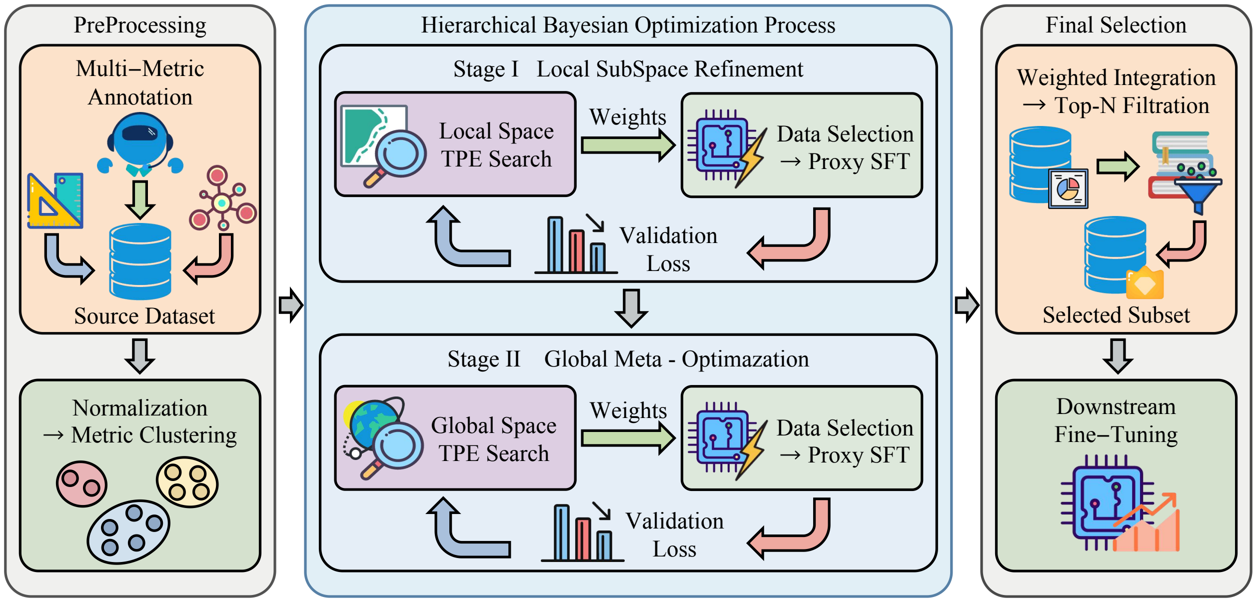

Figure 2: Overview of R-Select. The framework first performs multi-metric annotation, normalization, and Metric clustering; then executes two-stage hierarchical Bayesian optimization using proxy SFT; finally, the learned weights are applied for weighted integration and top-k selection before full supervised fine-tuning.

4.2 Hierarchical Bayesian Optimization Process

Addressing the non-differentiable nature of the ranking function $\mathcal{I}(\cdot)$ and the prohibitive computational cost of the bi-level objective $\mathcal{J}(\mathbf{w})$, we propose a tractable hierarchical decomposition strategy. This framework transforms the intractable high-dimensional optimization problem into a sequence of lower-dimensional subproblems, driven by the Tree-structured Parzen Estimator (TPE) [74]. By coupling dimensionality reduction with a budget-aware proxy evaluation protocol, our approach efficiently navigates the search space from local metric refinement to global semantic integration.

The Density-Based Solver. We employ TPE as our core derivative-free optimizer due to its robustness in handling non-convex landscapes and conditional dependencies. Unlike standard Gaussian Process (GP)-based Bayesian optimization, which models the objective function directly, TPE models the conditional probability of weight configurations $p(\mathbf{w} \mid y)$, where y denotes the observed validation loss. Specifically, TPE partitions historical trials $\mathcal{H} = \{(\mathbf{w}^{(t)}, y^{(t)})\}_{t=1}^{T}$ into two non-parametric densities based on a loss threshold y^* (typically the γ-quantile of observed losses):

$$p(\mathbf{w} \mid y) = \begin{cases} \ell(\mathbf{w}), & \text{if } y < y^* \quad \text{(promising regions)}, \\ g(\mathbf{w}), & \text{if } y \geq y^* \quad \text{(sub-optimal regions)}. \end{cases} \tag{3}$$

Candidate generation selects configurations that maximize the Expected Improvement (EI), which is proportional to the likelihood ratio $\ell(\mathbf{w})/g(\mathbf{w})$. This mechanism naturally balances the exploration of under-sampled regions with the exploitation of high-density regions associated with lower validation losses.

Hierarchical Decomposition Strategy. Directly optimizing $\mathbf{w} \in \mathbb{R}^F$ suffers from the *curse of dimensionality*, rendering standard BO sample-inefficient. Leveraging the metric clustering defined in Sec. 4.1, we decompose the optimization into two sequential stages. This effectively reduces the search complexity from $O(\exp(F))$ to $O(\exp(K) + \sum \exp(|C_k|))$, making the global optimum reachable under practical computational constraints.

Stage I: Intra-group Refinement. This stage targets the *intra-group* trade-offs within each cluster C_k. We formulate K independent optimization tasks to find local weight vectors $\mathbf{w}_k^{\text{loc}}$ on the probability simplex:

$$\min_{\mathbf{w}_k^{\text{loc}}} \mathcal{L}_{\text{val}}(\theta^*(\mathbf{w}_k^{\text{loc}}); \mathcal{D}_{\text{val}}) \quad \text{s.t.} \sum_{i \in C_k} w_{k,i}^{\text{loc}} = 1, \mathbf{w}_k^{\text{loc}} \succeq 0. \tag{4}$$

To accelerate convergence, we adopt a small-budget proxy strategy: candidates are evaluated using a reduced data budget $n_1 \ll N$ and the same lightweight proxy model. This acts as a cost-effective filter to identify dominant metrics within each semantic group. The optimized weights are then used to synthesize a robust composite signal $G_k(d) = (\mathbf{w}_k^{\text{loc}})^\top \tilde{\mathbf{f}}_{C_k}(d)$, effectively denoising the feature space.

Stage II: Inter-group Integration. The second stage performs *inter-group* integration in the reduced K-dimensional meta-space. To ensure scale consistency, we first normalize the signals as $\bar{G}_k(d) = \phi(G_k(d))$. We then optimize a meta-weight vector $\boldsymbol{\beta} \in [0, 1]^K$ to balance the contribution of each composite signal $\bar{G}_k$.

Recognizing that the meta-space is compact ($K \ll F$) yet critical for final performance, we switch to a larger-budget proxy evaluation with data budget n_2 ($n_1 < n_2 \ll N$). This minimizes the proxy gap variance, ensuring reliable global tuning. The final dense weight vector is reconstructed via the hierarchical product $w_i^* = \beta_k \cdot w_{k,i}^{\text{loc}}$, yielding a solution that captures both granular quality signals and high-level semantic balance.

4.3 Optimal Subset Curation and Training

Upon the convergence of the hierarchical optimization process, we obtain the optimal global weight vector $\mathbf{w}^*$. This vector represents the learned data selection policy that best balances diverse quality dimensions for the target task.

In the final deployment phase, we apply this policy to the entire candidate pool $\mathcal{D}$ (size M) to materialize the selection. Specifically, we compute the scalar quality score for every sample $d \in \mathcal{D}$ via the linear projection:

$$S_{\text{final}}(d) = (\mathbf{w}^*)^\top \hat{\mathbf{f}}(d). \tag{5}$$

We then construct the optimal training subset $\mathcal{D}^* \subset \mathcal{D}$ by selecting the top-N instances with the highest scores:

$$\mathcal{D}^* = \{d \in \mathcal{D} \mid \text{rank}(S_{\text{final}}(d)) \leq N\}. \tag{6}$$

Finally, the target LLM (which is independent of the proxy model used during search) is supervised fine-tuned on $\mathcal{D}^*$. Importantly, although the hierarchical optimization relies on a lightweight proxy model for tractable exploration, the learned weight vector $\mathbf{w}^*$ serves as a generally transferable data selection policy across downstream backbones. As a result, the final curation step is computationally trivial relative to full SFT training, while yielding substantial gains in data efficiency and model performance.

5 Experiments

In this section, we validate the effectiveness of our proposed R-Select framework through comprehensive comparisons, ablations, and scalability analyses across diverse domains and model scales.

5.1 Experimental Setup

5.1.1 R-Select Setting. We instantiate R-Select with a diverse suite of complementary data quality metrics spanning three broad categories of signals. These metrics jointly provide a heterogeneous view of data quality for selection. All metric annotations are generated using the OpenDataArena [5] toolkit, and full definitions and hyperparameters are provided in Appendix C. The candidate data pool is constructed from 21 heterogeneous instruction-tuning sources, covering a wide range of domains, styles, and reasoning patterns. Specifically, the pool includes datasets such as OpenThoughts3 [25], LIMO [82], Tulu-3-sft-mixture [37], among others. A complete list of data sources is summarized in Appendix A. For optimization and model selection, we employ a validation suite composed of well-established benchmarks, including GPQA [63], Omni-Math [22], MMLU [27], BigCodeBench [96], and MBPP [3]. These benchmarks are used exclusively for validation. Detailed validation set descriptions are provided in Appendix B. During the R-Select optimization, we use Qwen3-1.7B-Base [81] as a proxy model to enable efficient low- and medium-fidelity evaluations. The Tree-structured parzen estimator (TPE) optimizer is implemented using the Optuna [1] toolkit. Detailed configurations are reported in Appendix D.

5.1.2 Downstream Setting. We fine-tune on two backbones, Qwen3-8B-Base and Qwen2.5-7B-Base, using a unified SFT recipe implemented in LlamaFactory [92] with DeepSpeed ZeRO-3. All models are trained for 3 epochs with a cosine learning-rate schedule. Comprehensive training hyperparameters are provided in Appendix E. For evaluation, we adopt the OpenCompass [13] framework and exclude any benchmarks that appear in the validation set to avoid contamination. We report results on previously unseen benchmarks across four domains. For the General domain, we evaluate on IFEVAL [94], DROP [15] and MMLU-Pro [73]. For the Math domain, we include MATH500 [28], OlympiadBench [26], and AIME2024 [51]. For the Code domain, evaluation is performed on HumanEval [10], LCB (V5) [30], and HumanEval+ [44]. For the Reasoning domain, we report results on ARC-C [12], KOR-BENCH [50] and BBH [67]. More detailed evaluation protocols are provided in Appendix F.

5.2 Main Results

Table 1 and Table 2 present the main results of R-Select across two backbones and four evaluation domains. Based on these results, we draw three key findings regarding the effectiveness of R-Select:

1. R-Select is highly data-efficient and transferable. Table 1 compares R-Select with several widely used open-source high-quality SFT datasets across four domains (General, Math, Code, and Reasoning). Although R-Select is optimized using a much smaller proxy model and contains only 100k samples—substantially fewer than most baselines (ranging from 300k to nearly 1M)—it achieves the highest average performance on both Qwen2.5-7B and Qwen3-8B backbones. This demonstrates that a carefully selected compact subset can outperform much larger curated mixtures, and that the selection learned on a lightweight proxy model transfers effectively to larger base models. Across domains, R-Select shows particularly strong gains in math and reasoning while maintaining competitive general and code performance, suggesting a balanced rather than task-biased selection. The consistent improvements from Qwen2.5-7B to the stronger Qwen3-8B further indicate that the benefits of R-Select persist as model scale increases and generalize to previously unseen benchmarks.

2. Learnable weighting outperforms naive rules. Table 2 first contrasts R-Select with three rule-based multi-metric baselines: Random Sample, Uniform Average, and Top-K Intersection. Across both backbones, R-Select consistently achieves the highest average performance, indicating that naive strategies are insufficient for exploiting the complementary information in multiple metrics. In particular, Top-K Intersection performs reasonably on math but degrades noticeably on general and code tasks, suggesting that overly strict hard constraints discard valuable samples. Uniform Average, which treats all metrics equally, also underperforms R-Select, implying that different quality dimensions contribute unevenly to downstream performance. These results highlight the advantage of R-Select's learnable weighting scheme, which flexibly integrates heterogeneous metrics instead of relying on rigid heuristics.

3. Multi-metric coordination is necessary. The lower part of Table 2 compares R-Select with four representative single-metric selectors, including IFD [40], LLM-as-Judge Complexity, SkyworkRM [43], and Token-Length. We observe that all single-metric methods exhibit clear domain biases or instability: for example, Token-Length performs well on math but significantly lags in general tasks, while

Table 1: Comparison between our R-Select and representative open-source high-quality SFT datasets on overall four domain benchmarks. Best results within each section are shown in bold, and second-best results are underlined.

Dataset	Size	General			Math			Code			Reasoning			Avg
		Drop	IFEval	MMLU-P	MATH500	OLYMP	AIME'24	HE	HE+	LCB	ARC-C	BBH	K.B.	
Qwen2.5-7B-Base														
Qwen2.5-7B-Base [60]		68.3	35.5	44.2	50.2	35.9	6.7	77.4	43.3	8.2	36.6	69.5	33.3	45.8
LIMO [82]	817	73.5	53.3	54.7	66.8	34.9	4.6	**83.5**	59.8	17.6	90.9	55.6	48.3	53.6
Light-R1-SFT [75]	79k	79.4	38.5	46.3	88.0	60.2	38.3	45.7	40.2	3.2	78.0	72.7	52.3	53.6
SYNTHETIC-2-SFT [29]	105k	64.4	54.9	35.5	90.0	67.4	45.0	42.1	40.2	11.1	**93.2**	**81.1**	58.8	57.0
OpenThoughts [57]	114k	68.5	41.2	**55.6**	83.6	53.3	22.5	68.9	**68.9**	17.2	90.5	72.4	51.0	57.8
OmniThought [7]	365k	52.8	34.3	38.1	89.8	**68.1**	50.4	57.9	51.8	17.9	90.5	76.8	**58.9**	57.2
MiroMind [41]	719k	**82.3**	30.6	38.6	**91.6**	66.3	**55.0**	32.3	26.2	5.4	84.1	74.9	51.4	53.2
Tulu3-sft-mixture [37]	939k	62.3	**72.1**	46.0	55.5	29.0	4.4	79.1	66.7	12.3	80.7	61.9	48.5	51.5
R-Select	100k	65.7	46.6	46.9	86.6	64.4	31.7	65.9	63.6	**19.0**	87.5	66.8	52.2	**58.1**
Qwen3-8B-Base														
Qwen3-8B-Base [81]		71.5	45.9	56.2	79.6	47.2	6.7	82.9	34.2	16.9	37.3	78.1	46.6	50.3
LIMO [82]	817	72.0	52.3	49.8	69.0	31.3	12.5	81.1	62.2	13.6	83.4	48.7	45.7	51.8
Light-R1-SFT [75]	79k	83.4	46.8	57.7	92.6	69.7	54.6	81.7	65.9	14.7	92.2	84.5	60.4	67.0
SYNTHETIC-2-SFT [29]	105k	38.7	67.6	56.8	93.8	71.5	58.8	81.1	42.7	18.3	92.9	86.6	64.1	64.4
OpenThoughts [57]	114k	79.6	43.6	38.7	92.2	71.5	47.9	72.0	75.0	**31.5**	91.2	82.3	57.7	65.2
OmniThought [7]	365k	50.3	49.7	46.2	95.4	74.9	**67.9**	**91.5**	64.6	29.4	**93.9**	**86.8**	**65.0**	67.9
MiroMind [41]	719k	**85.0**	43.5	55.1	**96.8**	**77.0**	62.9	82.3	71.3	21.2	92.9	86.5	62.0	69.7
Tulu3-sft-mixture[37]	939k	56.3	**69.6**	44.7	56.4	34.4	4.2	72.0	62.8	15.1	85.4	64.7	47.0	51.0
R-Select	100k	81.8	56.4	**61.1**	94.4	71.1	56.7	81.1	**75.6**	28.0	92.9	80.6	59.1	**69.9**

model-based scorers tend to underperform on average despite excelling in specific cases. In contrast, R-Select consistently outperforms all single-metric baselines across both backbones and all four domains, validating the necessity of multi-metric coordination for high-quality SFT data construction.

5.3 Ablation Study

Table 3 ablates key design choices of R-Select by comparing two variants: (i) a flat optimization that directly searches over all 30 metrics in a single stage (R-Select_Flat), and (ii) a hierarchical variant that preserves the two-stage structure but selects only a single representative metric per cluster (R-Select_Exemplar). The results indicate that both the hierarchical structure and intra-cluster aggregation are essential to the effectiveness of R-Select.

Hierarchical vs. flat optimization. Compared with the flat baseline R-Select_Flat, the full two-stage hierarchical optimization consistently achieves higher average performance, with particularly notable gains on math benchmarks. This suggests that decomposing the high-dimensional search space into a coarse-to-fine process—first optimizing at the cluster level and then refining at the metric level—reduces optimization difficulty and mitigates noise in the weight space. In contrast, one-shot optimization over all metrics is more prone to suboptimal solutions and less capable of balancing competing quality dimensions.

Intra-cluster aggregation vs. exemplar selection. Results from R-Select_Exemplar reveal that selecting only the single highest-weighted metric per cluster leads to clear performance degradation, especially on math and reasoning tasks. This indicates that a single exemplar cannot fully capture the complementary signals within a cluster. By contrast, R-Select's weighted aggregation across multiple correlated but non-redundant metrics preserves richer information about sample quality, enabling more reliable selection.

Taken together, these ablations show that the hierarchical optimization structure of R-Select is not merely an engineering convenience but a critical design for effective data selection. Equally important, intra-cluster weighted aggregation is necessary to retain complementary information and avoid overly simplistic reductions to single metrics. Their combination underpins R-Select's strong and robust performance with a compact training set.

5.4 Robustness Analysis

We assess the robustness of R-Select along two dimensions: (i) the compressibility of the metric weights learned via hierarchical optimization, and (ii) the performance consistency of R-Select across different training data scales. Specifically, we evaluate three data sizes (10k, 50k, and 100k) and compare R-Select with Random Sample, Top-K Intersection, and Uniform Average.

1. Robustness to metric-count pruning. Table 3 reports results when varying the number of metrics used for final data selection while keeping the two-stage hierarchical optimization unchanged. When retaining only the top-10 or top-20 metrics (R-Select_10_Metrics and R-Select_20_Metrics), we observe a mild but consistent performance decrease compared to the full R-Select, with the top-20 variant generally outperforming the top-10 variant. This

Table 2: Comparison of R-Select against representative data selection baselines. Best and second-best are shown in bold and underlined.

Dataset	General	Math	Code	Reasoning	Avg
		Qwen2.5-7B-Base			
Qwen2.5-7B-Base [60]	49.3	30.9	42.9	46.4	45.8
Random Sample	51.0	60.3	38.1	63.9	53.3
Uniform Average	**55.1**	58.1	34.5	64.0	52.9
Top-K Intersection	51.6	<u>63.1</u>	38.7	<u>68.3</u>	<u>55.4</u>
IFD [40]	34.7	53.7	17.1	55.4	40.2
Complexity* [6]	44.4	53.9	19.0	60.8	44.5
SkyworkRM† [43]	42.8	53.7	18.3	61.7	44.1
Token Length	33.5	**69.3**	<u>47.4</u>	63.8	53.5
R-Select	<u>53.1</u>	61.0	**50.0**	**68.9**	**58.1**
		Qwen3-8B-Base			
Qwen3-8B-Base [60]	57.8	44.5	44.6	54	50.3
Random Sample	67.6	64.1	47.5	76.2	63.8
Uniform Average	<u>67.8</u>	71.0	46.2	**78.5**	<u>65.9</u>
Top-K Intersection	51.7	69.3	49.0	76.8	61.9
IFD [40]	46.1	71.2	49.8	68.3	58.9
Complexity* [6]	64.2	67.9	28.1	73.7	58.5
SkyworkRM† [43]	**69.9**	60.8	<u>53.7</u>	77.2	65.4
Token Length	53.0	**76.1**	46.3	**78.5**	63.5
R-Select	66.5	<u>74.1</u>	**61.6**	<u>77.6</u>	**69.9**

* Complexity based on LLM-as-Judge.
† Skywork-Reward-V2-Llama-3.1-8B-40M.

Table 3: Ablation on R-Select's hierarchical optimization and robustness to metric count. Results are reported on Qwen3-8B-Base with 50k samples; best and second-best are shown in bold and underlined.

Dataset	General	Math	Code	Reasoning	Avg
R-Select_Flat	**68.4**	<u>69.5</u>	43.0	**79.4**	<u>65.1</u>
R-Select_Exemplar	59.6	68.3	54.0	<u>77.3</u>	64.8
R-Select_10_Metrics	<u>65.3</u>	65.2	<u>54.5</u>	68.9	63.5
R-Select_20_Metrics	64.9	65.7	54.1	71.5	64.1
R-Select	64.1	**72.6**	**54.7**	75.9	<u>66.8</u>

suggests that the metric weights learned by our hierarchical procedure are reasonably compressible: a small subset of high-weight metrics preserves much of the effectiveness, while the full 30-metric set still provides the most comprehensive quality signals and yields the best overall performance. In practice, this indicates that R-Select can flexibly trade off computation for performance when needed.

2. Performance across different data scales. Figure 3 compares R-Select with baselines at 10k, 50k, and 100k samples. Across all three scales, R-Select consistently outperforms Random Sample, Top-K Intersection, and Uniform Average. While all methods benefit from increased data size, the magnitude of improvement differs markedly across strategies. In the low-data regime (10k–50k), R-Select exhibits the largest margin over baselines, demonstrating superior data efficiency when training data is limited. At 100k

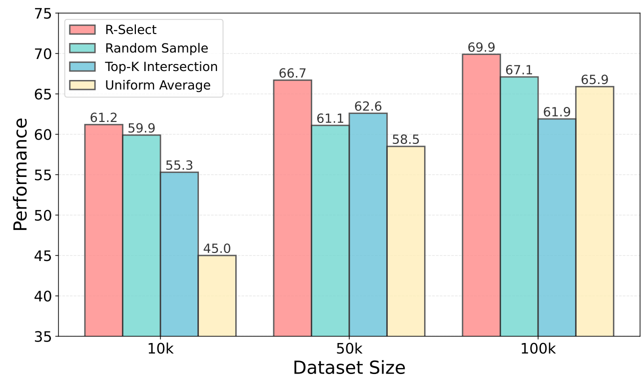

Figure 3: Scaling trends of R-Select v.s. other baselines under varying selection budgets. All experiments are conducted using Qwen3-8B-Base on a fixed dataset of 50k samples.

samples, R-Select continues to improve and maintains a clear performance lead over all baselines. These results indicate that R-Select not only excels in data-scarce settings but also remains effective as the dataset scales.

Overall, these experiments demonstrate that R-Select is robust along both the metric dimension and the data-scale dimension: its learned weights can be effectively compressed with limited performance loss, and its advantages persist across a wide range of training sizes, highlighting strong data efficiency and scalability.

6 Interpreting R-Select via Visualization

This section provides a visual interpretation of R-Select's underlying selection mechanism. By analyzing learned weight patterns and subset diversity, we demonstrate how R-Select effectively prioritizes high-quality data while maintaining extensive semantic coverage.

6.1 Learned Weight Distribution Analysis

We examine the weight map learned by R-Select as shown in Fig. 4. Rather than being uniform, the learned weights display a highly structured and sparse pattern, concentrating on a small set of high-utility signals. This allows us to interpret how different dimensions of data quality contribute to effective SFT data selection.

6.1.1 Cluster-level insights. The cluster-level meta-weights β_k indicate an emphasis on learning difficulty and distributional properties. C_4 (Learning Difficulty & Distribution, $\beta_4 = 0.644$) dominates the weight map, suggesting that the selected data are primarily shaped by sample learnability, information gain, and distributional stability rather than surface-level textual quality. This is consistent with the objective of SFT, where challenging yet learnable samples tend to be most beneficial. C_1 (Structural Complexity & Richness, $\beta_1 = 0.187$) emerges as the second most influential dimension. While C_4 captures how difficult a sample is to learn, C_1 characterizes how that difficulty is organized through syntactic depth, token entropy, and sample scale, providing structural constraints on the selected data. C_6 (Educational Value & Stylistic Quality, $\beta_6 = 0.092$) plays a complementary calibration role by preserving factual reliability and expressive clarity. In contrast, C_3 (Model Alignment Preference)

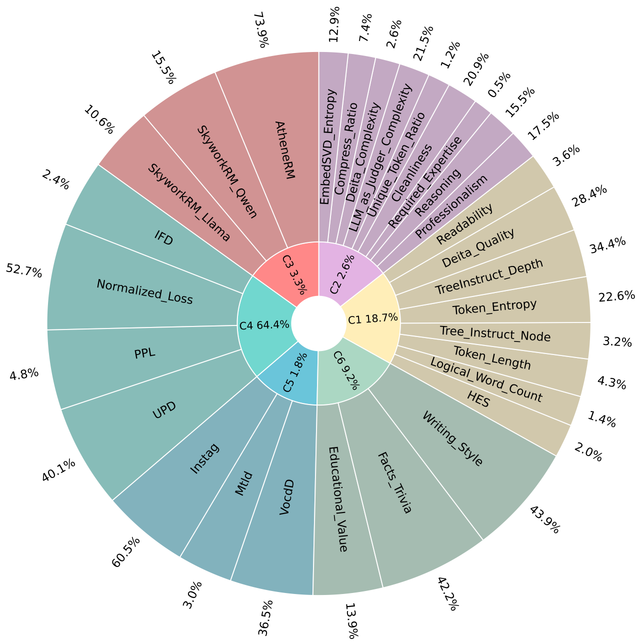

Figure 4: Weight map learned by R-Select. Brighter cells indicate higher importance assigned to a metric during data selection, revealing both intra-cluster sparsity and inter-cluster heterogeneity

receives relatively small weight and mainly serves as a lightweight alignment safeguard rather than a primary selection driver. C_2 and C_5 contribute marginally, indicating substantial overlap with the dominant clusters for our evaluated tasks.

6.1.2 Intra-cluster patterns. Within each cluster, the local weights $w_{k,i}^{loc}$ consistently exhibit a "sparse head" pattern: a few metrics dominate while others receive much lower weight (Fig. 4). In C_4, Normalized_Loss ($w_{4,i}^{loc} = 0.527$) and UPD (0.401) are the primary contributors, jointly emphasizing sample learnability and distributional coherence, whereas traditional difficulty measures such as IFD and PPL are substantially down-weighted. In C_1, TreeInstruct_Depth, Deita_Quality, and Token_Entropy concentrate most of the weight, highlighting hierarchical structure, learned quality estimation, and information density, while Logical_Word_Count and HES receive minimal weight due to overlapping signals. C_6 is mainly driven by Writing_Style and Facts_Trivia, with Educational_Value playing a secondary role, whereas C_3 is largely represented by AtheneRM, indicating that a single stable reward-model signal is sufficient for lightweight alignment control.

Taken together, the learned weight map suggests a compact trade-off in which selection is primarily guided by learnability (C_4), constrained by structural coherence (C_1), and lightly calibrated by textual value and alignment (C_6/C_3).

6.2 Data Diversity and Composition Analysis

A key concern for any data selection method is whether performance gains come at the cost of reduced diversity. We examine

Figure 5: Comparison of semantic coverage before and after applying R-Select.

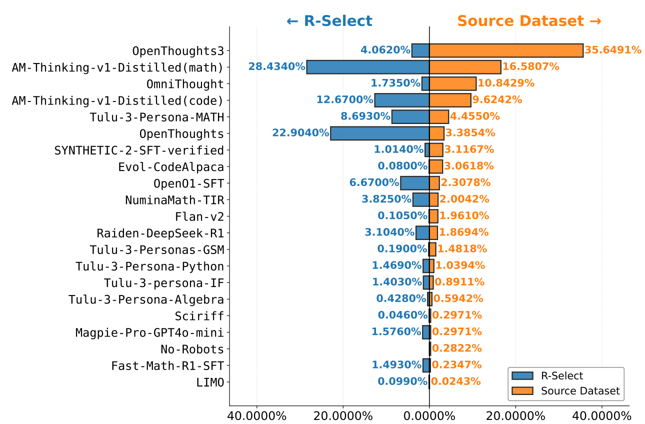

Figure 6: Comparison of source distributions between the original data pool and R-Select.

this from two complementary perspectives: (i) semantic coverage in embedding space, and (ii) changes in source composition after selection. Together, these analyses assess whether the 100k samples selected by R-Select retain the breadth of the original data pool while emphasizing higher-quality regions.

6.2.1 Semantic diversity in embedding space. To compare semantic coverage, we randomly sample 10k instances from both the original Source Dataset and the R-Select subset, encode them using Qwen3-Embedding-8B [90], and visualize the representations with t-SNE [70] as shown in Fig. 5. Visually, the two distributions exhibit comparable global spread, indicating that R-Select does not collapse the data into a narrow semantic region. Instead, the selected samples largely occupy the same major manifolds as the full pool. At a finer granularity, high-density areas in the R-Select plot appear slightly more concentrated, suggesting that the method preferentially retains coherent regions of the space while pruning isolated or noisy points. Crucially, we observe no evidence of mode collapse or dominance by a single domain. Overall, the embedding analysis supports that the 100k subset preserves the semantic diversity of the original corpus despite its much smaller size.

6.2.2 Source composition and selection bias. Because the Source Dataset aggregates corpora with different prior performance, we analyze R-Select through two lenses: (i) changes in source composition and (ii) per-source selection rate (the fraction of each source retained), which better reveals the method's intrinsic preference than raw counts alone. As shown in Fig. 6, R-Select does not preserve original proportions. Instead, it systematically shifts mass toward higher-performing reasoning and math sources such as AM-Thinking-v1-Distilled (math), OpenThoughts, and AM-Thinking-v1-Distilled (code), while heavily compressing very large synthetic corpora (notably OpenThoughts3). This indicates a clear preference for per-sample value rather than dataset scale. The selection-rate view is more revealing. High-performing sources exhibit substantially higher retention rates than their size alone would predict, whereas lower-scoring or highly templated datasets are pruned more aggressively. Several small but strong sources (e.g., LIMO and Raiden-DeepSeek-R1) retain a meaningful share of samples, suggesting that R-Select recognizes high-quality niches rather than defaulting to large corpora.

7 Discussion

In this section we examine key aspects of proxy-based data selection with R-Select, including transferability to target models, validation set configuration, metric robustness, and computational efficiency.

7.1 Proxy-to-Target Transferability

A central question in low-resource data selection is whether weights learned on a small proxy model generalize to larger target models. We conduct experiments to evaluate this transfer, showing that the policy learned on a 1.7B Qwen3-Base proxy model correlates strongly with downstream performance on 1.7B, 4B, and 8B Qwen3 models (Spearman's $\rho > 0.89$), indicating robust transfer across model sizes [39]. We further examine how architectural differences affect transfer. Using the same 1.7B Qwen3-Base proxy model, we evaluate downstream models of similar size (7–8B) but different architectures, including Qwen2.5, Qwen3, and Llama3.1. The results show that transfer effectiveness decreases as the architectural gap increases, although proxy-learned policies still provide meaningful guidance [11].

7.2 Validation Set Configuration

In proxy-based optimization, the configuration of the validation set plays an important role in guiding data selection. To provide reliable optimization signals, we construct the validation objective from representative datasets spanning multiple domains. We also study the effect of validation subset size on selection outcomes. Across 10 weight configurations, rankings of validation loss remain consistent for subsets of 2.5k, 5k, 10k, and 15k samples (Kendall's $\tau > 0.82$). These results show that even small proxy subsets provide informative guidance, allowing smaller subsets to be used to improve computational efficiency without sacrificing effectiveness [89].

7.3 Metric Selection and Robustness

R-Select leverages a set of 30 data-quality metrics covering multiple dimensions of interest. The set is flexible: the utility of each metric is learned automatically, and additional metrics can be incorporated as needed. As shown in Table 3, comparisons between the full metric set and pruned subsets show that using the complete set consistently improves selection performance by capturing both strong metrics and complementary information from weaker or redundant ones, producing similar outcomes across different metric choices, qualities, and redundancies.

7.4 Computational Efficiency

Proxy-based optimization introduces additional computation but remains efficient relative to full downstream training. In our experiments, the proxy search requires 113 GPU hours, compared to 384 GPU hours for a single 100k-sample downstream SFT run, representing approximately 23% of total compute. This fraction decreases as the downstream training scale increases. Furthermore, compared with simpler baselines and multi-metric approaches that rely on manual tuning or repeated full SFT, R-Select provides a practical balance of computational cost and performance.

8 Conclusion

This paper presents R-Select, a scalable framework for data-centric selection in LLM post-training. Instead of relying on ad hoc heuristics or costly black-box scorers, we formalize data selection as a structured weight optimization problem over a multi-dimensional quality space. By integrating 30 complementary indicators into a unified scoring function, R-Select provides a systematic way to model the trade-offs among diversity, difficulty, and general linguistic properties of SFT data. Methodologically, we introduce a Hierarchical Optimization strategy that decomposes high-dimensional selection into manageable sub-problems. We first cluster correlated metrics into functional groups and perform intra-group refinement to capture local structure, then conduct inter-group integration to balance global objectives. This divide-and-conquer design improves optimization stability and efficiency while preserving performance gains. Empirically, R-Select consistently outperforms both heuristic baselines and strong model-based selectors across multiple downstream tasks. More broadly, our results highlight that structured optimization of data composition, as opposed to simply increasing model scale or scoring complexity, can play a decisive role in improving post-training outcomes for large language models.

9 GenAI Disclosure

We employed large language models (GPT-5 and GPT-4o) as auxiliary tools during paper writing. Their usage was confined to non-technical writing support, including grammar checking, stylistic adjustments, and improvements in clarity and fluency. All technical ideas, dataset construction, experimental design, and result analysis originate solely from the authors. The use of LLMs did not contribute to the scientific content of this work and served only to facilitate more fluent and polished writing.

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A Source Datasets

Our source data pool aggregates over 3.4 million samples from diverse publicly available SFT datasets covering multiple domains and paradigms, including reasoning, code, math, and instruction-following. Key datasets include OpenThoughts3 [25], AM-Thinking-v1-Distilled (math and code) [32], OmniThought [7], OpenThoughts [25], SYNTHETIC-2-SFT-verified [29], Open01-SFT [77], Raiden-DeepSeek-R1 [65], Fast-Math-R1-SFT [83], Tulu-3-Persona-MATH [19], NuminaMath-TIR [33], LIMO [82], Tulu-3-Persona-Python [17], Tulu-3-Persona-IF [18], Tulu-3-Persona-Algebra [16], Magpie-Pro-10K-GPT4o-mini [80], SciRIFF [71], Evol-CodeAlpaca [49], FLAN-v2 [46], No-Robots [62],Tulu-3-Persona-GSM [20].

B Validation Dataset

For evaluating and optimizing our data selection strategies, we use a validation dataset comprising 2,255 samples drawn from several publicly available benchmarks. Specifically, we sample 600 instances from Omni-Math [22], 546 from GPQA [63], 553 from MMLU [27], 301 from BigCodeBench [96], and 255 from MBPP [3]. This set serves as a reliable performance proxy for our framework.

C Metric Details

We employ a comprehensive suite of evaluation metrics to capture complementary dimensions of SFT data quality. The metrics used in our framework include: MTLD [52], Token_Entropy [95], Token_Length, TreeInstruct_Node [91], TreeInstruct_Depth [91], Unique_Token_Ratio [56], VOCD-D [52], Logical_Word_Count, Compress_Ratio, EmbedSVD_Entropy [84], AtheneRM [21], Cleanliness [95], Deita_Complexity [45], Deita_Quality [45], HES [2], IFD [40], Instag [48], NormLoss [66], PPL [31], Professionalism [95], Writing_Style [76], Required_Expertise [76], Facts_Trivia [76], Educational_Value [76], Readability [95], Reasoning [95], SkyworkRM_Llama [43], SkyworkRM_Qwen [43], UPD [86], and LLM_as_Judge_Complexity. For more detailed descriptions and implementation details, please refer to our GitHub repository.

D Implementation Details

We implement the Hierarchical Bayesian Optimization framework for R-Select using a two-stage search with proxy data budgets of $n_1 = 2.5$k and $n_2 = 5$k samples. The optimization is performed with the Optuna toolkit using a Tree-structured Parzen Estimator (TPE) sampler, initialized with structured warm-up trials drawn from a Dirichlet prior and supplemented by boundary points for broad simplex coverage. Early stopping is applied with patience of 200 and delta of 1e−4, and the total search budget is capped at 1000 trials. All searches use a fixed random seed of 42, and weights are defined on $[0, 1]$ with linear normalization onto the probability simplex. The proxy model is trained with DeepSpeed using the ds_z3_config setting, default template, cutoff length of 32,768 tokens, 16 preprocessing workers, packing enabled, per-device batch size of 1, gradient accumulation of 4 steps, learning rate 5e−5, Liger kernel enabled, 3 training epochs, cosine learning rate scheduler, and warmup ratio 0.1.

E Downstream Training Details

We adopt a standardized SFT setup for all downstream experiments using DeepSpeed ZeRO-3 optimization (ds_z3_config) and a consistent training recipe. The models are trained with the default template, cutoff length of 32,768 tokens, 16 preprocessing workers, and packing enabled. Each device uses a batch size of 2 with gradient accumulation of 2 steps. The learning rate is set to 5e−5 with the Liger kernel enabled. Training runs for 3 epochs with a cosine learning rate scheduler and a warmup ratio of 0.1.

F Downstream Evaluation Details

We use OpenCompass as the evaluation framework. Benchmarks span four key domains: General, Math, Code, and Reasoning. For the General domain, we evaluate DROP using xVerify-9B-C in a 3-shot setting with accuracy; IFEval using IFEvaluator in 0-shot; and MMLU-PRO using xVerify-9B-C in 5-shot. In Math, benchmarks include OlympiadBenchMath, MATH-500, and AIME'24 using xVerify-9B-C in 0-shot, measuring accuracy (for AIME'24, average over 8 runs). For Code, we evaluate HumanEval using HumanEvalEvaluator, HumanEval+ using HumanEvalPlusEvaluator, and LiveCodeBench (v5) using LCBCGgenerationEvaluator, all in 0-shot with pass@1. In Reasoning, benchmarks include ARC_C, BBH, and KOR-Bench using xVerify-9B-C in 0-shot with accuracy as the metric.